

Requirements Table

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Problem Statement 49 : Webhook Ingestion and Retry Mechanism Hub

So basically this system is like a webhook gateway. It takes events coming from outside third party services, keeps a log of the payloads, sends them to our internal services and if some delivery fails it keeps retrying with exponential backoff .. If it still fails after all tries then it puts the event into a dead letter queue so nothing gets lost.

Actors : Integration Developer, System Operator, Third Party Service (the outside event sender)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR001	Functional	The system shall ingest incoming webhooks, forward the payload to the target endpoint and auto retry failed deliveries using exponential backoff (1s,2s,4s,8s).	High	Pass : failed delivery gets retried and logged. Fail : webhook dropped with no dead letter save.	This is the main job of the whole gateway.
FR002	Functional	The system shall log every incoming webhook payload along with its source, timestamp and the delivery status.	High	Pass : each webhook makes one log row with source, time, status. Fail : a webhook has no log.	Needed for audit and to debug when stuff breaks.
FR003	Functional	The system shall move a webhook into the Dead Letter Queue once the retry attempts are finished	High	Pass : after 4th fail the webhook is in DLQ with payload intact. Fail : it just gets thrown away.	So that no event is lost when the endpoint is down.

		(after 4 failed tries).			
FR004	Functional	The system shall let the System Operator view and replay the webhooks that are sitting in the Dead Letter Queue.	Medium	Pass : operator lists and replays, success removes it from DLQ. Fail : cant list or replay.	Lets us recover manually once the service is back up.
FR005	Functional	The system should allow an Integration Developer to register and configure the target endpoints (url, retry limit, timeout).	Medium	Pass : new endpoint gets the forwarded webhooks with its config. Fail : config is ignored.	Makes the gateway reusable for many services.
NFR001	Non Functional (Performance)	The ingestion endpoint shall buffer incoming webhooks with 99.999% reliability and handle throughput upto 2500 requests per second.	High	Pass : benchmark under peak load hits the target latency and reliability.	Peak load resilience stops events getting lost.
NFR002	Non Functional (Security)	The system shall authenticate every incoming webhook by checking a HMAC signature and reject any request whose	High	Pass : valid signature accepted, invalid one returns 401 and is not processed.	Keeps fake or spoofed payloads out of the internal services.

		signature is missing or wrong.			
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